Step 1: Data Acquisition and Initial Exploration

First, access and download the two primary datasets:

- **Bitcoin Market Sentiment Dataset**
- **Historical Trader Data from Hyperliquid**

Use the provided Google Drive links to get the data. Once downloaded, load the datasets into a data analysis environment like a Jupyter Notebook with Python.

Step 2: Data Cleaning and Preparation

Before analysis, prepare the data:

- **Handle Missing Values:** Check for any missing data points in both datasets and decide on a strategy for handling them (e.g., imputation or removal).
- **Date/Time Alignment:** The core of the project is linking trader performance to market sentiment. The `Date` column from the sentiment data and the `time` column from the trader data need to be in a consistent, standardized format. Convert them to a common datetime format to enable accurate merging.
- **Data Structure:** Examine the columns in both datasets. The sentiment data has `Date` and `Classification` (Fear/Greed). The trader data includes various columns like `closedPnL`, `leverage`, `side`, and `symbol`.

Step 3: Feature Engineering and Analysis

Create new features and perform the core analysis to explore the relationship between sentiment and performance.

- **Join the Datasets:** Merge the cleaned trader data with the sentiment data based on the aligned date/time columns. This will allow you to associate each trade with the corresponding market sentiment (`Fear` or `Greed`).
- **Define Performance Metrics:** Use the `closedPnL` column to calculate key performance indicators (KPIs) for each trader and across different sentiment periods. Consider metrics like average PnL, win rate (the percentage of trades with a positive `closedPnL`), or cumulative PnL.
- **Grouped Analysis:** Segment the trader data by sentiment (`Fear` vs. `Greed`) and perform a comparative analysis. For example, compare the average PnL of all trades that occurred during `Fear` days to those that occurred during `Greed` days.

Step 4: Uncovering Patterns and Visualization

Look for hidden patterns and trends to provide deeper insights.

- **Identify Top Performers:** Find the most profitable traders (`account`) based on their cumulative `closedPnL` and analyze their trading behavior.
- **Behavioral Analysis:** Investigate how different trader behaviors correlate with sentiment.
  - **Leverage:** Do traders use higher leverage on `Greed` days?
  - **Trade Side:** Is there a correlation between the `side` (buy/sell) of a trade and the market sentiment?
- **Visualize the Findings:** Use charts and graphs to make your findings clear and compelling. Examples include:
  - A bar chart comparing average PnL during `Fear` vs. `Greed` periods.
  - A time-series plot of a top trader's cumulative PnL overlaid with the market sentiment index.

Step 5: Final Insights and Recommendations

Synthesize your findings and create a report.

- **Summarize Key Findings:** Clearly state the relationship you found between market sentiment and trader performance. For instance, "We found that while most traders are more profitable on 'Greed' days, a select group of top traders successfully profit from 'Fear' days by taking short positions."
- **Propose Trading Strategies:** Based on your insights, propose actionable trading strategies. A strategy might be: "When the market sentiment is 'Fear', high-leverage short positions on BTC have a higher probability of being profitable for experienced traders."